

EuroSAT Land-Use Classification

Custom CNN (from scratch) vs. ResNet50 (transfer learning)

MSc in Artificial Intelligence & Visual Computing (AIVC)
University of West Attica
Assignment (Ergasia) 2026 — Image Classification

A comparative study of two deep-learning approaches for land-use / land-cover classification on 27,000 Sentinel-2 satellite tiles across 10 classes.

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1. Problem description

EuroSAT is a benchmark dataset of Sentinel-2 satellite images for land-use / land-cover (LULC) classification. It contains 27,000 RGB images of 64×64 pixels, evenly spread across 10 classes: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River and SeaLake. The goal of this project is to train, analyse and compare at least two different deep-learning models for this classification task, evaluating their performance, capabilities and limitations.

2. The Idea (methodology)

To contrast two of the approaches suggested in the assignment, we implement and compare:

Model	Approach	Rationale
Custom CNN	Trained from scratch	Compact VGG-style net (4 conv blocks) built for 64×64 tiles — a lightweight baseline.
ResNet50	Transfer learning + fine-tuning	ImageNet-pretrained backbone with a new classifier head, fully fine-tuned.

Pipeline (identical for both models, for a fair comparison):

1. Data — download EuroSAT (Zenodo) and build a reproducible stratified 70/15/15 train/validation/test split (seed = 42) → 18,900 / 4,050 / 4,050 images.
2. Augmentation — random horizontal/vertical flips and $\pm 15^\circ$ rotation (satellite tiles are orientation-agnostic). The CNN uses native 64×64 with EuroSAT normalization; ResNet50 resizes to 224×224 with ImageNet normalization.
3. Training — CrossEntropyLoss, Adam, ReduceLROnPlateau scheduler; best checkpoint chosen by validation accuracy.
4. Hyperparameter optimization — Optuna (5 trials) tunes learning rate, dropout, batch size and optimizer.
5. Evaluation — accuracy + per-class precision/recall/F1, confusion matrix and a sample-prediction grid on the held-out test set.

3. Analysis — overall results

Summary — test-set performance

Model	Approach	Test acc.	Macro F1	Trainable params	Epoch time
Custom CNN	from scratch	0.9632	0.9621	457,738 ($\approx 0.46\text{M}$)	~ 5.3 s
ResNet50	transfer learning + fine-tuning	0.9874	0.9870	23,528,522 ($\approx 23.5\text{M}$)	~ 125 s

ResNet50 reaches the highest test accuracy (98.7%), about +2.4 points over the from-scratch CNN (96.3%), and converges to >0.98 validation accuracy in a single fine-tuning epoch — confirming the value of ImageNet transfer learning on satellite imagery. The Custom CNN remains a strong, $\sim 51\times$ lighter baseline.

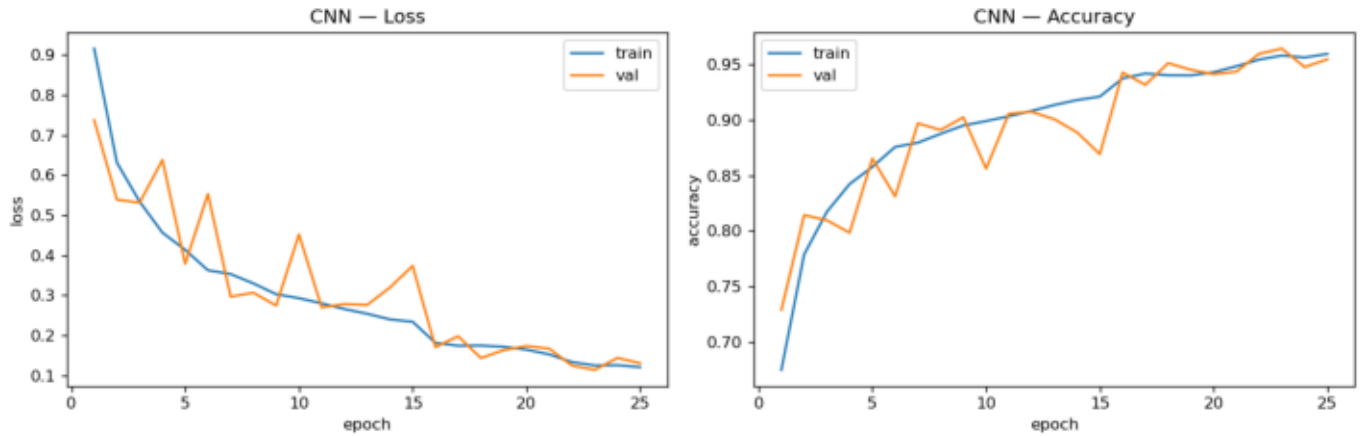
Optuna best configuration (5 trials each)

Model	lr	dropout	batch	optimizer	val acc (5-epoch trial)
Custom CNN	5.61e-04	0.580	32	adam	0.849
ResNet50	1.53e-03	0.219	32	sgd	0.978

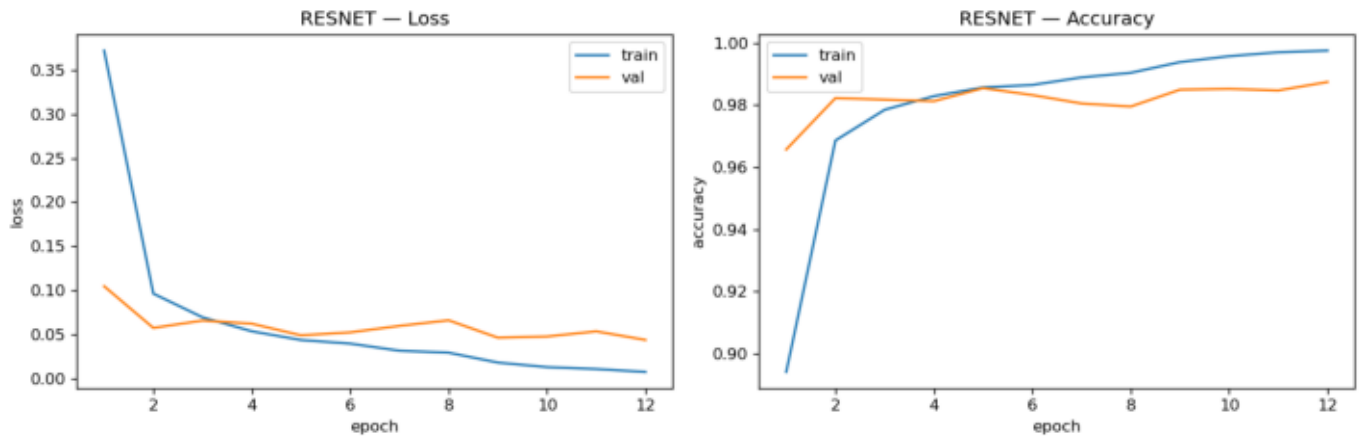
The search optimizes validation accuracy over short 5-epoch trials. Full per-trial tables are committed under `results/optuna_cnn_trials.csv` and `results/optuna_resnet_trials.csv`.

Training curves (Loss / Accuracy)

Custom CNN



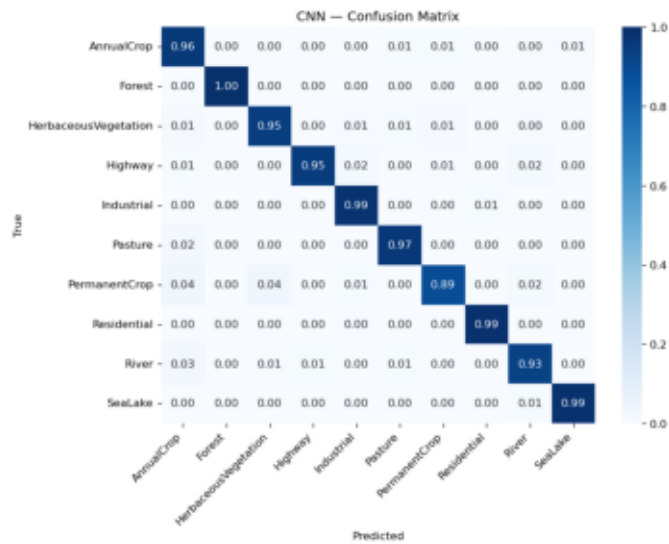
ResNet50



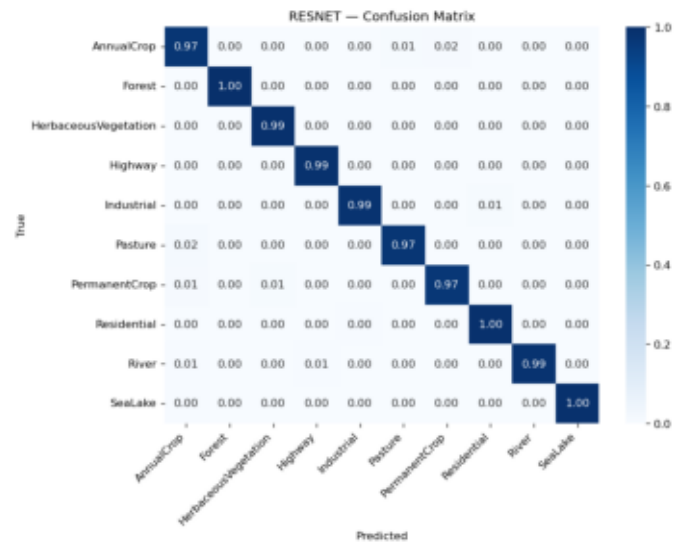
Per-epoch training and validation loss and accuracy. ResNet50 starts near its ceiling thanks to pretrained features; the CNN improves steadily over the full schedule.

Confusion matrices (test set)

Custom CNN



ResNet50



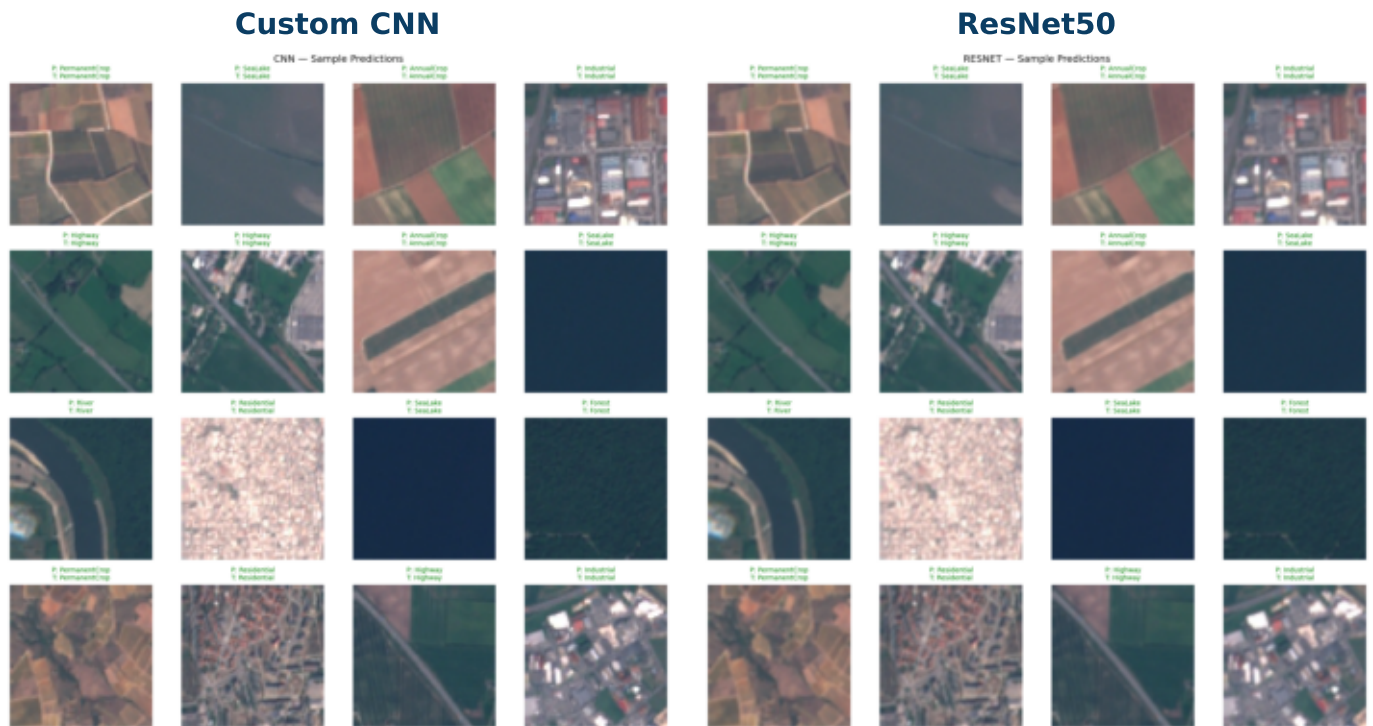
Counts of true vs. predicted classes. ResNet50's off-diagonal mass is much lighter, concentrated on the crop/vegetation classes.

Per-class metrics (test set)

Class	CNN P	CNN R	CNN F1	ResNet P	ResNet R	ResNet F1
AnnualCrop	0.908	0.962	0.934	0.967	0.973	0.970
Forest	0.989	1.000	0.994	0.996	1.000	0.998
HerbaceousVegetation	0.960	0.949	0.954	0.989	0.993	0.991
Highway	0.981	0.949	0.965	0.984	0.992	0.988
Industrial	0.959	0.987	0.972	0.989	0.987	0.988
Pasture	0.954	0.967	0.960	0.983	0.973	0.978
PermanentCrop	0.962	0.888	0.924	0.976	0.971	0.973
Residential	0.989	0.993	0.991	0.991	0.996	0.993
River	0.938	0.933	0.936	0.997	0.987	0.992
SeaLake	0.991	0.989	0.990	1.000	0.996	0.998
macro avg	0.963	0.962	0.962	0.987	0.987	0.987

Both models are strongest on visually distinct classes (Forest, SeaLake, Residential) and weakest on the spectrally similar crop/vegetation classes. For the Custom CNN, PermanentCrop (recall 0.89), River (0.93) and HerbaceousVegetation (0.95) are the hardest; ResNet50 is much flatter, keeping every class ≥ 0.97 recall.

Sample predictions (green = correct, red = wrong)



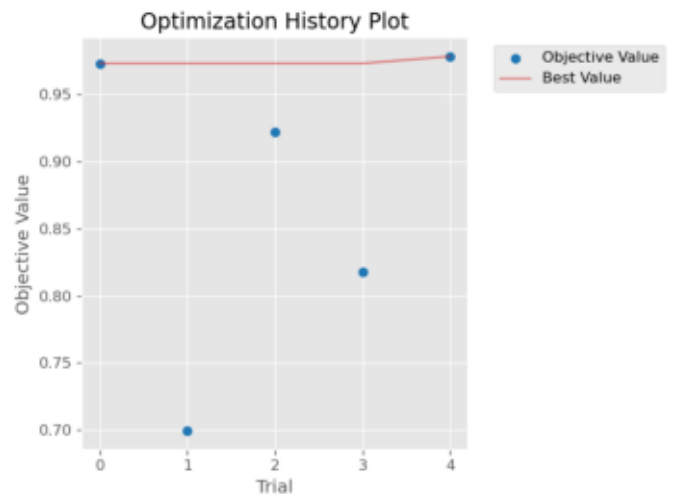
Random held-out test tiles with predicted labels.

Optuna hyperparameter search history

Custom CNN



ResNet50



Validation accuracy per trial and the running best over the 5-trial search.

4. Comparison, capabilities & limitations

Custom CNN (from scratch)

- + Lightweight (457K params, $\sim 51\times$ fewer than ResNet50), ~ 5 s/epoch, low memory — runs on modest hardware.
- + Operates on native 64×64 tiles (no upscaling needed).
- + Still reaches a strong 96.3% test accuracy from scratch.
- Lower accuracy ceiling; must learn all visual features from limited data.
- More sensitive to augmentation / regularization choices.

ResNet50 (transfer learning + fine-tuning)

- + Highest accuracy (98.7% test, +2.4 pts over the CNN); ImageNet features transfer well to satellite imagery.
- + Converges fast — >0.98 validation accuracy after a single fine-tuning epoch.
- ~ 23.5 M parameters; $\sim 24\times$ slower per epoch (~ 125 s), heavier memory and inference cost.
- Requires upscaling $64\times 64 \rightarrow 224\times 224$ and ImageNet normalization.

Shared failure mode

Both models confuse the same semantically/visually similar land-cover classes — PermanentCrop $\leftrightarrow$ AnnualCrop and HerbaceousVegetation $\leftrightarrow$ Pasture — which is expected for low-resolution 64×64 satellite tiles where texture cues are limited (see confusion matrices).

Conclusion

If accuracy is the priority and a GPU is available, ResNet50 with transfer learning is the clear choice. Where compute, memory or latency are constrained, the Custom CNN delivers 96%+ accuracy at a fraction of the cost — a favourable accuracy/efficiency trade-off. All randomness is seeded (SEED = 42), so the split and training are reproducible.